Another allotrophic form of carbon is graphene. It has a single planar sheet of sp^2 hybridised carbon atoms that are densely packed in a honeycomb crystal lattice.

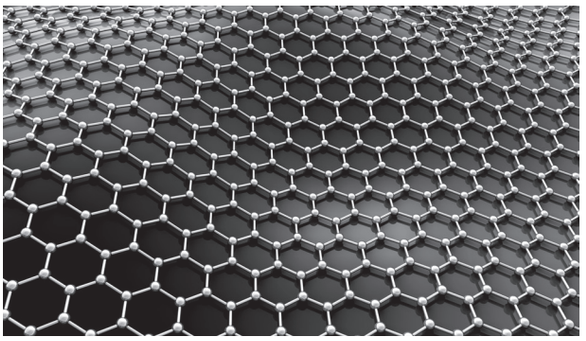


Figure 2.8 Structure of graphene

2.3.5 Carbon monoxide [CO]:

Preparation:

Carbon monoxide can be prepared by the reaction of carbon with limited amount of oxygen.

$$2C + O_2 \longrightarrow 2CO$$

On industrial scale carbon monoxide is produced by the reaction of carbon with air. The carbon monoxide formed will contain nitrogen gas also and the mixture of nitrogen and carbon monoxide is called producer gas.

$$2C + O_2/N_2\ (air) \longrightarrow \underset{\text{Producers Gas}}{2CO + N_2}$$

The producer gas is then passed through a solution of copper(I)chloride under pressure which results in the formation of $CuCl(CO).2H_2O$. At reduced pressures this solution releases the pure carbon monoxide.

Pure carbon monoxide is prepared by warming methanoic acid with concentrated sulphuric acid which acts as a dehydrating agent.

$$HCOOH + H_2SO_4 \longrightarrow CO + H_2SO_4 \cdot H_2O$$

Properties

It is a colourless, odourless, and poisonous gas. It is slightly soluble in water.

It burns in air with a blue flame forming carbon dioxide.

$$2CO + O_2 \longrightarrow 2CO_2$$

When carbon monoxide is treated with chlorine in presence of light or charcoal, it forms a poisonous gas carbonyl chloride, which is also known as phosgene. It is used in the synthesis of isocyanates.

$$CO + Cl_2 \longrightarrow COCl_2$$

Carbon monoxide acts as a strong reducing agent.

$$3CO + Fe_2O_3 \longrightarrow 2Fe + 3CO_2$$

Under high temperature and pressure a mixture of carbon monoxide and hydrogen (synthetic gas or syn gas) gives methanol.

$$CO + 2H_2 \longrightarrow CH_3OH$$

In oxo process, ethene is mixed with carbon monoxide and hydrogen gas to produce propanal.

XII_U2-P-Block.indd 43 04-01-2021 12:48:12